

Equivariant Systems Theory and Observer Design for Autonomous Systems

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Outline

- 1 Practical session: observer design on $SO(3)$

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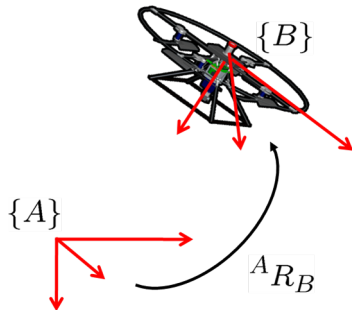
- 1 Practical session: observer design on $SO(3)$

Orientation between two frames

Attitude can be represented as rotation matrices in the special orthogonal group

$$\text{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^\top R = I, \det(R) = 1\}$$

Alternative representations are $q \in \mathbb{R}^4$ unit quaternions, or as roll-pitch-yaw Euler angles (ψ, θ, ϕ) .



Attitude kinematics

Left invariant kinematics

$$\dot{R} = R\Omega_{\times}$$

$\Omega \in \{B\}$ is the rigid body angular velocity measured in the body-fixed-frame.

$$\Omega_{\times} = \begin{pmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{pmatrix} \in \mathfrak{so}(3)$$

The angular velocity map $\times$ is the matrix representation of the vector product. That is $\Omega_{\times} v = \Omega \times v$ for any vector $v \in \mathbb{R}^3$.

Motivation: Attitude estimation for your Smartphone

3 kinds of sensors present:

- 3-axis accelerometers
- 3-axis rate gyro
- 3-axis magnetometers

Rate gyrometers: measure angular velocity in the body-fixed frame

$$\Omega_y = \Omega + b + \mu_b$$

Accelerometers Readings: measure the linear accelerations applied to the mobile in the body-fixed frame

$$\dot{v} = RAcc + ge_3; \quad Acc_y = Acc + \mu_a$$

If $\dot{v}$ is low, $Acc_y \approx -gR^\top e_3 + \mu_a \Rightarrow R_y^\top e_3 = -\frac{Acc_y}{||Acc_y||}$

Attitude observer

System kinematics

$$\dot{R} = R\Omega_{\times}, \quad y_i = R^T \dot{y}_i$$

Attitude observer

System kinematics

$$\dot{R} = R\Omega_{\times}, \quad y_i = R^T \dot{\hat{y}}_i$$

Observer on the group system

$$\dot{\hat{R}} = \hat{R}\hat{\Omega}_{\times}, \quad \hat{y}_i = \hat{R}^T \dot{\hat{y}}_i, \quad \hat{\Omega} = \Omega + k_P \sum_{i=1}^2 y_i \times \hat{y}_i$$

Attitude observer

System kinematics

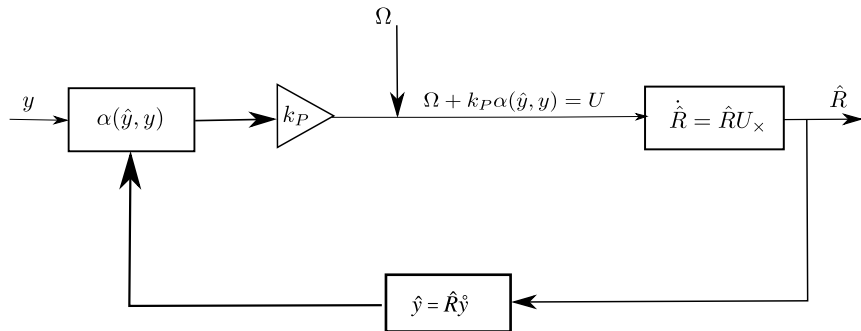
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$$\Rightarrow \hat{R}_{k+1} = \hat{R}_k \exp(\hat{\Omega}_{\times}^k T), \quad \hat{R}(0) = \hat{R}_0 \text{ (e.g. } \hat{R}_0 = I)$$

Block Diagram of Observer



Attitude Estimation using Smartphone Sensors

Aim:

Get practical experience in implementing algorithms and gain tuning.

Visualize and compare noisy and filtered signals.

We will collect data from an Android smartphone using the application "Sensor Fusion"

Then we will process these data in Matlab to reconstruct the attitude of the phone.

Attitude Estimation using Smartphone Sensors

Matlab files:

Create a new folder for your project e.g. "observer"

Get the matlab files from your instructor and copy them in the folder

read_log_script.m: reads the .txt file generated by the application and extracts matlab arrays. This script is automatically called in the following main scripts

obs_exp_ske....m: constructs the attitude of the phone directly from accelero and magneto measurements without filtering and leaves places for you to complete the code

Attitude Estimation using Smartphone Sensors

Android phone

Download the application "Sensor Fusion" on your android phone
It's a free application created at Linköping University and available
on GooglePlay (<http://goo.gl/0qNyU>)

Make sure you choose the correct application (check the app's
icon) If you use another application the `read_log_script.m` file
should be adapted



Attitude Estimation using Smartphone Sensors

Logging Data:

Run the application

You can start noticing the axis convention of the application by visualizing a sensor's measurement in "Select Sensor"

Choose the magnetometer, put your phone on an horizontal plane in the direction of the magnetic north ($y = 0$) and write down the x-y-z measurements (will be useful in initializing the matlab code) click on "Log Data" on the main page

Select the following sensors "Accelerometer", "Gyroscope", "Magnetic Field"

Deselect "Stream" on the top of screen, and select "Log" instead

Attitude Estimation using Smartphone Sensors

Logging Data:

When you are ready to record your phone's motion click on "Start", then "Stop" when you finish.

Make sure to create an interesting motion of your phone e.g. Start with the phone on an horizontal plane then do pure rolling (around the x-axis) followed by a pure pitching (around y-axis) then a yawing (around z-axis)

The App should have created a .txt file on your phone. Send it to your PC, copy it in your project's directory and rename it to "sensorLog.txt"

Attitude Estimation using Smartphone Sensors

Iphone

- 1 install "Phyphox" application and run it.
- 2 click on "+" at the top right and choose "Add simple experiment". Choose a title for the experiment, and set the Sensor rate to '100' Hz. Select the following sensors:" Accelerometer," " Gyroscope," and" Magnetic Field," and save the settings.
- 3 scroll to "Simple custom experiments" and click on the created experiment. When you are ready to record your phone's motion, click "Start" to record and then "Stop" when you finish.
- 4 export the data in .csv format, send it to your PC and copy it into *data_iphone* folder in your project's directory.
- 5 ensure the files are named: "Accelerometer.csv", "Gyroscope.csv", "Magnetometer.csv" and run the Matlab script *read_log_script_iphone*
- 6 run your estimator using your script

Attitude Estimation using Smartphone Sensors

Direct attitude reconstruction:

Run the "att_construct.m" script

Notice the effect of noisy measurement

Explicit filter with ignoring the bias on gyro measurements

Define a matlab function that compute the skew symmetric matrix associated with any vector in $\mathbb{R}^3$: $Sk(v): \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$

$$Sk(v) = \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix} \in \mathfrak{so}(3)$$

Define a matlab function that compute the "exponential" of any 3-vector: $ExpSk(M): \mathbb{R}^3 \rightarrow \mathbb{R}^{3 \times 3}$

$$ExpSk(M) = I + \bar{M} \sin(|M|) + \bar{M}^2 (1 - \cos(|M|)), \quad \bar{M} = Sk(M/|M|)$$

Attitude Estimation using Smartphone Sensors

Build the observer that we already designed in case where the bias is assumed to be zero

- ① Run your code and tune your gain until you extract reasonable signals and check that the innovation as defined the measure of the magnetic field affects the estimation of pitch and roll angles,
- ② Modify the code to decouple the measure of the accelerometers from the magnetic field measure, by replacing $\dot{y}_2$ by $\dot{y}_2' = \dot{y}_1 \times \dot{y}_2$ that leads to $y_2' = R^\top (\dot{y}_1 \times \dot{y}_2) = (R^\top \dot{y}_1 \times R^\top \dot{y}_2) = y_1 \times y_2$.
- ③ Modify the code in order to deal with the bias and think of choosing small gains for the integrator
- ④ Compare with the previous cases.